

EDUCATION

Bachelors of Science in **Professional Physics**. Montana State University, 2018. **3.95 GPA**

WORK EXPERIENCE

Tonic.ai

Senior Frontend Engineer

Aug 2024 - Present

- Teamed up with a backend engineer to **implement role-based access control (RBAC)** to enable compliance with government security standards and **unlock a contract valued at \$800K**
- **Improved auditability** of PII results by adding in entity preview and correction. This allowed the team to **more effectively market to enterprise customers**
- Proactively **maintained and patched dozens of frontend dependencies** reducing security vulnerabilities and ensuring compliance with enterprise security policies

Emotive.io — (2021 - 2024)

Lead Software Engineer

September 2023 - June 2024

- Worked with finance team to lead a project automating the import + upload + charging of monthly billing spreadsheets. Uploading + parsing + validating a csv using Django **prevented the org from doing 100s of hours of manual, error-prone tasks**. Uploading the documents to S3 ensured auditability
- Implemented a TCPA litigator list screening feature to allow brands to filter out enrollments by those who would be looking to sue them, effectively **saving brands \$1500 per litigator** identified
- Updated and implemented a frontend technical screening process which allowed the org to identify cost-effective contractors and ultimately **Reduced the team's labor costs by 50-60%**
- Guided a team of contractors based overseas. Worked as a liaison between calls with brands and engineers to **translate requests into technical requirements**. Provided code review + coordinated discussions surrounding best practices

Senior Software Engineer

November 2021 - September 2023

- Transitioned portions of a Django based monolith to a Flask + FastAPI based microservice architecture to ensure scalability while **maintaining a cyclomatic complexity of 5 or lower**. This made it possible for teams to function more independently and increased the org's velocity
- Refactored Django systems to **reduce query times by 99.9%** by introducing covering indices and lateral joins into our PostgreSQL database. This allowed the database to be downsized which lowered operational costs.
- Rewrote legacy React components to **remove unnecessary global state** + abstractions allowing for a **75% reduction in total lines of code** for a microfrontend
- **Lead the company's "Frontend Guild"** to educate and elevate the organization's frontend engineers. We explored topics like the transition to React 18, when to use various state management solutions, as well as a general best practice consensus.

15Five Inc — (2019 - 2021)

Engineering Manager

March 2021 - Oct 2021

- Led a team of 4 engineers tasked with maintaining and **improving the core 15Five Check-in feature** of our app.
- Cultivated a healthy atmosphere by encouraging direct reports to bring their full selves to work through regular 1-on-1's, performance reviews, and role clarity discussions which led to **100% retention of the engineers on my team**.
- **Collaborated with designers, product managers, mobile engineers, and other teams** while participating in customer calls to incorporate feedback, build new features, and integrate other teams' work into the Check-in.

Technical Lead

Oct 2020 - Oct 2021

- **Developed end to end solutions** for all feature work. We typically began by establishing scope before writing and presenting an implementation plan and tickets. During the development cycle I would **write code and review 100% of the relevant pull requests**.
- Implemented TypeScript adoption at an org level. I worked with several other developers to **enforce static typing on 100% of our React components**. This was part of the effort that allowed our DevOps team to embrace CI/CD and **decrease cycle time by 50% org-wide**.
- Architected and built an extensible framework to simplify app-wide rich text editing. I began by investigating existing usage in order to determine the proper abstractions before **removing 1000s of lines of duplicated code and several abstraction layers** from our codebase.

Frontend Engineer

Oct 2019 - Oct 2020

- Built OKR tracking feature into existing app infrastructure. This feature has **grown into our most adopted feature**. In order to implement this, we had to **integrate custom React hooks and functional components into legacy jQuery** business logic.

Triple Tree LLC

Software Engineer

Mar 2019 - Aug 2019

- Built a veteran transition program with the COMMIT Foundation. **Implemented OAuth2** to ensure data security and developed a video streaming and recommendation engine by **transforming client designs into React + Redux components**.
- Researched, Designed, & Implemented app wide **email automation logic**, canary releases on AWS, a **video recommendation engine** built in NodeJS

The Business Garage

Junior Software Engineer

Feb 2018 - Mar 2019

- Built the backend using Django and **managed deployments for a web app with 50K users**. We experienced viral growth and **implemented redis caching to reduce load times by 60%**. Prior to the growth, I ensured scalability by developing distributed load test infrastructure through AWS.
- Researched, Designed, & Implemented **push notifications** through a redis queue, **content caching** to reduce db stress, **distributed load testing** on AWS

LIGO Scientific Collaboration

Researcher

2013, 2016 - 2018

- Wrote python applications + received grant funding to develop deep neural networks using Tensorflow which **modeled neutron stars** to test Einstein's theory of gravity.

OTHER PROJECTS, EXPERIENCE & SKILLS

Skills

- **Frontend:** TypeScript, JavaScript, Webpack, HTML, CSS, Sass, React, Redux, React Hook Form, React Final Form, TanStack (query, table, virtual, etc), jQuery, D3, Tailwind, Bootstrap, some (but limited) angular + vue
- **Backend:** Python, Django, Django REST Framework, Django Channels, Celery, Flask, FastAPI, Node.js, Express.js, AdonisJS, Hapijs, Redis, PostgreSQL, CouchDB (+ some MongoDB), SQLite, MariaDB, RethinkDB
- **DevOps/Cloud:** Docker, Docker Compose, AWS (EKS, Lambda, Gateway, SSM, RDS, Cloudwatch, S3, EBS among others), Kubernetes, Heroku
- **Data Science:** Spacy, TensorFlow, Pandas, Jupyter, SciPy, SKLearn, NLTK, Matplotlib, Seaborn, Bokeh, Keras
- **Other:** Linux, Bash, GNU Make, Git, Mathematica, vim

Open source contributions—  github.com/jpribyl &  johnpribyl.com

- Wrote, published, and maintain react-hook-dragula and react-hook-mathjax which have received around 20 stars and 300 downloads per week. I have also **contributed to React, Django**, vim-test, and Freqtrade.
- Forked, fixed a deprecation warning, and released action-docker-layer-caching when unable to reach the original maintainer.

Teaching & Coaching

- A combined 8 years of experience coaching soccer and tutoring math and physics up to a college level and have worked with ages ranging from 5 to 50 years and **subjects ranging from arithmetic to quantum mechanics.**

Grand Canyon Rafting

- Successfully led a 3 week, unguided raft trip down 280 river miles through the canyon. I planned all the logistics for 16 people.